The Imperfective Paradox in Large Language Models

大语言模型中的未完成体悖论

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Abstract

摘要

Do Large Language Models (LLMs) genuinely grasp the compositional semantics of events, or do they rely on surface-level probabilistic heuristics? We investigate the Imperfective Paradox, a logical phenomenon where the past progressive aspect entails event realization for activities (e.g., running □ ran) but not for accomplishments (e.g., building □ built). We introduce IMPERFECTIVENLI, a diagnostic dataset designed to probe this distinction across diverse semantic classes. Evaluating state-of-the-art open-weight models, we uncover a pervasive Teleological Bias: models systematically hallucinate completion for goal-oriented events, even overriding explicit textual cancellation. Prompting interventions partially reduce this bias but trigger a calibration crisis, causing models to incorrectly reject valid entailments for atelic verbs. Representational analyses further show that while internal embeddings often distinguish progressive from simple past forms, inference decisions are dominated by strong priors about goal attainment. Taken together, our findings indicate that these current open-weight LLMs operate as predictive narrative engines rather than faithful logical reasoners, and that resolving aspectual inference requires moving beyond prompting toward structurally grounded alignment. [1]

大型语言模型（LLMs）是否真正理解了事件的组合语义，还是仅仅依赖于表面的概率启发式方法？我们研究了 “未完成体悖论”，这是一种逻辑现象，其中过去进行时态蕴含了活动事件的实现（例如，running □ ran），但不适用于成就事件（例如，building □ built）。我们引入了 IMPERFECTIVENLI 数据集，该数据集旨在跨多种语义类别探测这种区别。通过评估最先进的开放权重模型，我们发现了一种普遍存在的目的论偏差：模型会系统性地对目标导向事件进行虚假完成，甚至会覆盖明确的文本取消信息。提示干预在一定程度上减少了这种偏差，但引发了校准危机，导致模型错误地拒绝了非谓语动词的有效蕴含。表示分析进一步表明，尽管内部嵌入通常能够区分进行时与一般过去时形式，但推理决策往往受到关于目标达成的强先验知识的影响。总体而言，我们的研究结果表明，这些当前的开放权重 LLM 更像是预测叙事引擎，而非忠实的逻辑推理者，并且解决时态推理需要超越提示方法，转向结构化的对齐。[1]

1 Introduction

1 引言

Large Language Models (LLMs) have demonstrated remarkable proficiency in natural language inference (NLI) and complex reasoning tasks (Brown et al., 2020; Touvron et al., 2023). Despite these advancements, questions remain regarding the robustness of their reasoning: do models genuinely grasp the compositional semantics of events, or do they rely on surface-level probabilistic heuristics (Bender et al., 2021; Dziri et al., 2023)? This

大型语言模型（LLMs）在自然语言推理（NLI）和复杂推理任务中表现出色（Brown 等，2020；Touvron 等，2023）。尽管取得了这些进展，关于其推理的鲁棒性仍存在疑问：模型是否真正理解了事件的组合语义，还是仅仅依赖于表面的概率启发式方法（Bender 等，2021；Dziri 等，2023）？这

Class	Examples (Premise □ Hypothesis)	Entail?
Telic	P: The carpenter was building a gazebo.H: The carpenter built a gazebo.	□ No
Atelic	P: The boy was running in the park.H: The boy ran in the park.	□ Yes

类	例子（前提 □ 假设）	蕴含?
目标性（Telic）	P: 工人正在建造一个凉亭。H: 工人建造了一个凉亭。	□ 不
非及物动词	P: 男孩在公园里跑步。H: 男孩在公园里跑。	□ 是

Table 1: Examples for the Imperfective Paradox. Telic verbs do not entail completion in the progressive form.

表 1：未完成体悖论的例子。结果性动词在进行体形式中不蕴含完成。

distinction is critical when processing temporal events. A faithful reasoning agent must distinguish between an action intended (or in progress) and an action realized.

区分在处理时间事件时是至关重要的。一个忠实的推理代理必须区分一个意图（或正在进行）的行为和一个已实现的行为。

Consider the following sentence: “The carpenter was building a gazebo”, as shown in Table 1. Logically, this description of an ongoing process does not entail that the gazebo was ever finished. The construction could have been abandoned or interrupted. This lack of entailment for telic (goal-oriented) events is known in formal semantics as the Imperfective Paradox (Dowty, 1977, 1979). In contrast, for atelic verbs like “run”, the process itself constitutes the event (Vendler, 1957; Bennett and Partee, 1978; Smith, 1997): “The boy was running” entails “The boy ran”.

考虑以下句子：“木匠正在建造一个凉亭”，如表 1 所示。从逻辑上讲，这种对正在进行过程的描述并不意味着凉亭曾经被建造完成。建造过程可能被放弃或中断。这种对目标导向事件（telic events）缺乏必然性的现象在形式语义学中被称为 “未完成体悖论”（Dowty, 1977, 1979）。相比之下，对于非目标导向动词（atelic verbs）如 “run”，过程本身构成了事件（Vendler, 1957; Bennett and Partee, 1978; Smith, 1997）：“男孩正在跑步” 蕴含 “男孩跑过来了”。

This phenomenon is grounded in the theory of (Lexical) Aspect, which categorizes events by their telicity (boundedness) (Vendler, 1957; Comrie, 1976; Leiss, 1992). Unlike atelic activities where the process implies the event (e.g., running), telic accomplishments possess a distinct culmination point that separates the process from the result (Moens and Steedman, 1988). Therefore, the progressive aspect acts as a logical filter: it entails realization for activities but suspends it for accomplishments. A robust reasoner must respect this structural boundary, distinguishing between the occurrence of a process and the attainment of a goal.

这一现象基于（词法）方面理论，该理论根据事件的完成性（有界性）对事件进行分类（Vendler, 1957; Comrie, 1976; Leiss, 1992）。与无完成性活动（如跑步）不同，其中过程隐含事件，完成性成就具有一个明确的终点，将过程与结果区分开来（Moens 和 Steedman, 1988）。因此，进行体（progressive aspect）作为一种逻辑过滤器：它对活动蕴含实现，而对成就则暂停实现。一个强大的推理者必须尊重这一结构边界，区分过程的出现与目标的达成。

While humans intuitively navigate these aspectual distinctions, we hypothesize that LLMs exhibit a Teleological Bias: a tendency to assume that

虽然人类直觉地导航这些方面差异，我们假设 LLMs 表现出一种目的论偏差：一种倾向于假设的倾向

goal-directed actions inevitably lead to their successful completion. This bias likely stems from the “reporting bias” in pre-training corpora (Gordon and Van Durme, 2013), where narratives typically mention goals only when they are relevant to the outcome, often implying success (Grice, 1975). Consequently, models may hallucinate event completion based on statistical likelihood rather than logical entailment, effectively conflating the intent with the result (Farquhar et al., 2024).

目标导向的行为不可避免地导致其成功完成。这种偏差很可能源于预训练语料库中的 “报告偏差” (Gordon 和 Van Durme，2013)，其中叙述通常只在目标与结果相关时才提及目标，往往暗示成功 (Grice，1975)。因此，模型可能会根据统计可能性而非逻辑蕴含来虚构事件的完成，实际上将意图与结果混为一谈 (Farquhar 等，2024)。

Prior work in NLP has largely focused on aspect classification or modeling, training models to label verbs as stative, dynamic, telic, or atelic (see Friedrich et al., 2023). However, high classification accuracy does not guarantee robust inference. It remains unexplored whether generative models can apply these aspectual properties to resist fallacious entailments in zero-shot reasoning scenarios.

自然语言处理（NLP）领域的先前研究在很大程度上集中于方面分类或建模，训练模型以将动词标记为静态、动态、目的性或非目的性（参见 Friedrich 等，2023）。然而，高分类准确率并不能保证稳健的推理。尚未探索生成模型是否可以应用这些方面属性来抵抗零样本推理场景中的谬误蕴含。

To bridge this gap, we introduce IMPERFECTIVENLI, a diagnostic dataset to probe the Imperfective Paradox. We construct minimal pairs across four logical conditions, contrasting interrupted and ambiguous contexts for both telic and atelic verbs. We evaluate state-of-the-art open-weight models and uncover following critical insights:

为了弥合这一差距，我们引入了 IMPERFECTIVENLI，一个用于探测未完成体悖论的诊断数据集。我们构建了四个逻辑条件下的最小对立对，对比了终止性动词和非终止性动词在中断和模糊语境下的差异。我们评估了最先进的开放权重模型，并发现了以下关键见解：

1) Teleological Bias Dominates Model Inference: Models exhibit a pervasive tendency to hallucinate completion for goal-oriented events. Crucially, this bias is robust enough to override explicit textual negation (e.g., predicting completion even when an action is interrupted), revealing a fundamental failure in contextual fidelity.
2) Prompting Interventions Trigger a Calibration Crisis: Strategies like Chain-of-Thought or forced counterfactual non-endpoints can reduce hallucinated completions, but often overcorrect, causing models to incorrectly reject valid entailments for atelic activities. Models struggle to adapt reasoning dynamically based on event type.
3) Representation and Reasoning Are Dissociated: We find that the bias is not representational; embeddings successfully distinguish process (progressive form) from result (simple past form). Instead, inference is dominated by prior expectations about goal completion. The failure is likely driven by predictive priors during decoding, rather than an inherent inability of Transformer architectures to encode the aspectual distinction.

1) 目的论偏见主导模型推理：模型表现出一种普遍的倾向，即对目标导向事件进行虚构补全。关键的是，这种偏见足够强大，以至于能够覆盖显式的文本否定（例如，即使某个动作被中断，模型仍会预测其完成），揭示了模型在上下文保真度方面的根本性失败。
2) 提示干预引发校准危机：诸如思维链或强制反事实非终点等策略可以减少虚构补全，但常常过度纠正，导致模型错误地拒绝了非谓语活动的有效蕴含。模型在根据事件类型动态调整推理方面存在困难。
3) 表示与推理是分离的：我们发现这种偏见并非源于表示层面；嵌入能够成功地区分过程（进行时形式）与结果（一般过去时形式）。相反，推理主要受到关于目标完成的先验预期的影响。这种失败很可能是由解码过程中的预测先验所驱动，而不是 Transformer 架构本身无法编码时态区分的固有缺陷。

Ultimately, our findings underscore that current LLMs struggle to resolve the Imperfective Paradox, operating as probabilistic predictors of narrative outcomes rather than faithful logical reasoners. Bridging this gap requires not just better prompting, but a deeper alignment between model behavior and the formal logic of event semantics.

最终，我们的研究结果表明，当前的大型语言模型（LLMs）在解决非完形悖论方面存在困难，它们更像是对叙事结果的概率预测者，而不是忠实的逻辑推理者。弥合这一差距不仅需要更好的提示方法，还需要模型行为与事件语义的形式逻辑之间更深层次的对齐。

2 Background

2 背景

2.1 Vendler's Aspectual Classes: Telicity

2.1 Vendler 的时态范畴：完成性

Vendler (1957) categorized verbs into four aspectual classes (activity, accomplishment, achievement, state) based on their temporal properties. In this study, we focus on the distinction between Activities and Accomplishments, defined by the property of telicity, whether an event involves an inherent endpoint. Telicity often correlates with perfective interpretations, which present an eventuality from a holistic perspective (Filip, 2008).

Vendler (1957) 根据动词的时间特性，将动词分为四个时态类别（活动、成就、完成、状态）。在本研究中，我们关注活动（Activity）与成就（Accomplishment）之间的区别，这一区别由目的性（telicity）这一属性界定，即事件是否包含内在的终点。目的性通常与完成体（perfective）解释相关，从整体视角呈现事件的必然性（Filip, 2008）。

- Activities (Atelic): Events with no inherent endpoint (e.g., “run”, “swim”, “push a cart”). They are homogeneous; any part of the process counts as an instance of the same event.

- Accomplishments (Telic): Events that proceed toward a specific terminus or goal (e.g., “build a house”, “write a letter”). Such events are only completed and receive canonical perfective interpretations, when the endpoint is reached.

- **活动（非事态性）**：没有固有终点的事件（例如：“跑步”，“游泳”，“推车”）。它们是同质的；过程中的任何部分都可以视为同一事件的实例。

- **成就（事态性）**：朝着特定终点或目标进行的事件（例如：“盖房子”，“写信”）。此类事件只有在达到终点时才被视为完成，并获得标准的完成体解释。

2.2 The Imperfective Paradox

2.2 不完成体悖论

Semantically, in English, the progressive form has imperfective reading, while the simple past form has perfective reading (Comrie, 1976; Smith, 1997). The Imperfective Paradox arises when we analyze the logical entailment between the progressive form (Imperfective) and the simple past form (Perfective) of these verb classes. Let ϕ be an event predicate and $PROG(\phi)$ denote its progressive form.

从语义上看，在英语中，进行体具有未完成体的读法，而简单过去时则具有完成体的读法（Comrie, 1976; Smith, 1997）。当我们将这些动词类的进行体（未完成体）与简单过去时（完成体）之间的逻辑蕴含关系进行分析时，就会出现未完成体悖论（The Imperfective Paradox）。设 ϕ 为一个事件谓词，$PROG(\phi)$ 表示其进行体形式。

For Atelic predicates (Activities), the entailment holds:

对于非结果谓词（活动），蕴含关系成立：

$$PROG(\text{run}) \vDash \text{run}$$

If x was running, then it is true that x ran. This is often referred to as the Sub-interval Property: if an activity holds true for an interval, it holds true for any sub-interval (Bennett and Partee, 1978).

如果 x 在运行，那么 x 确实运行过。这通常被称为子区间性质：如果一个活动在一个区间内成立，那么它在该区间的任何子区间内也成立（Bennett 和 Partee，1978）。

For Telic predicates (Accomplishments), the entailment fails:

对于 Telic 谓词（成就类动词），蕴含关系不成立：

$$PROG(\text{build a house}) \nvDash \text{build a house}$$

If x was building a house, it is NOT necessarily true that x built a house.

如果 x 在建一座房子，那么 x 建了一座房子这一说法并不一定成立。

The paradox highlights that the progressive form of an accomplishment describes a process leading to a result, but does not assert the result itself (Parsons, 1990). Dowty (1977) formalized this using possible worlds semantics, suggesting that $PROG(\phi)$ is true if ϕ completes in all "inertia worlds" (worlds where the course of events proceeds without interruption). However, the real world is not necessarily an inertia world and interruptions occur.

这个悖论指出，成就事件的进行体形式描述了一个导致结果的过程，但并不断言该结果本身（Parsons，1990）。Dowty（1977）使用可能世界语义学对这一点进行了形式化，认为如果 ϕ 在所有 "惯性世界"（即事件过程不间断进行的世界）中完成，则 $PROG(\phi)$ 为真。然而，现实世界并不一定是惯性世界，中断现象是存在的。

2.3 Related Work: Interpreting and Modeling "Aspect" in NLP

2.3 相关工作：自然语言处理中 "方面" 的解释与建模

Computational modeling of aspect has evolved from feature-engineering approaches to representation learning (Friedrich et al., 2023). Siegel and McKeown (2000) pioneered the use of supervised machine learning for aspectual classification. Subsequent work extensively explored semi-supervised approaches, combining linguistic and distributional features to predict stativity, duration, and situation entity types, while also establishing essential annotated datasets (Friedrich and Palmer, 2014; Friedrich and Pinkal, 2015; Friedrich et al., 2016; Friedrich and Gateva, 2017). Similar methodologies were applied to German verbs (Hermes et al., 2015) and multilingual morphosyntactic annotation (Ramm et al., 2017).

方面计算建模已从特征工程方法发展到表示学习（Friedrich 等，2023）。Siegel 和 McKeown（2000）率先将监督机器学习应用于方面分类。随后的研究广泛探索了半监督方法，结合语言学和分布特征来预测静止性、持续时间和情境实体类型，同时建立了重要的标注数据集（Friedrich 和 Palmer，2014；Friedrich 和 Pinkal，2015；Friedrich 等，2016；Friedrich 和 Gateva，2017）。类似的方法被应用于德语动词（Hermes 等，2015）和多语言形态句法标注（Ramm 等，2017）。

Moving toward compositional semantics, Kober et al. (2020) introduced an NLI-based dataset to model aspect in context. Later work showed that transformer models capture some aspectual cues but rely heavily on surface features such as tense and word order (Metheniti et al., 2022). More recently, Ma (2024) found that large-scale LLMs still struggle with lexical aspect identification. Closely related are two studies on the Imperfective Paradox (Pruś et al., 2024, 2025), which focus primarily on modeling human judgments of progressive versus simple past forms, treating LLM behavior as a secondary comparison. As such, neither study provides a systematic diagnostic evaluation of LLM aspectual inference. Our work directly addresses this gap, offering the first controlled investigation of whether LLMs respect the Imperfective Paradox or instead hallucinate event completion by conflating goal-directed processes with their result states.

向组合语义学方向发展，Kober 等人（2020）引入了一个基于自然语言推理（NLI）的数据集，以在上下文中建模方面。后来的研究表明，转换器模型能够捕捉一些方面线索，但严重依赖于诸如时态和词序等表层特征（Metheniti 等人，2022）。最近，Ma（2024）发现，大规模的大型语言模型（LLMs）在词法方面识别上仍然存在困难。与之密切相关的是两项关于未完成体悖论（Pruś 等人，2024，2025）的研究，这些研究主要关注建模人类对进行时与简单过去时形式的判断，将大型语言模型的行为作为次要的比较对象。因此，这两项研究都没有对大型语言模型的方面推理进行系统的诊断评估。我们的工作直接解决了这一差距，首次进行了受控研究，探讨大型语言模型是否尊重未完成体悖论，或者相反，通过将目标导向过程与其结果状态混淆，而错误地完成事件。

3 Dataset - IMPERFECTIVENLI

3 数据集 - IMPERFECTIVENLI

To evaluate LLMs' ability to reason about the imperfective paradox and teleological aspectual semantics, we introduce IMPERFECTIVENLI, a diagnostic NLI dataset in English.

为了评估大型语言模型在处理未完成体悖论和目的论事态语义方面的能力，我们引入了 IMPERFECTIVENLI，这是一个用于诊断自然语言推理的英文数据集。

Dataset Construction with Different Verbs.

使用不同动词的数据集构建。

The construction of IMPERFECTIVENLI employs a template-based generation approach to ensure syntactic uniformity while isolating semantic reasoning. We curated a balanced lexicon of verbs based on Vendler's aspectual classification:

IMPERFECTIVENLI 的构建采用基于模板的生成方法，以确保句法的一致性，同时隔离语义推理。我们根据 Vendler 的时态分类，精心整理了一个平衡的动词词汇表：

- 100 Telic Verbs (Accomplishments): Predicates involving a natural endpoint or result state, where the process is distinct from the culmination (e.g., build, write, fix, paint, bake).

- 100 Atelic Verbs (Activities): Predicates describing homogeneous processes without a necessary endpoint, where the process itself constitutes the event (e.g., run, swim, wander, play, vibrate).

- 100 有终点动词（完成体）：涉及自然终点或结果状态的谓词，其中过程与最终结果有所区别（例如：建造、写作、修理、绘画、烘烤）。

- 100 无终点动词（活动体）：描述同质过程且没有必要终点的谓词，其中过程本身即构成事件（例如：跑步、游泳、徘徊、玩耍、振动）。

For each verb, we generated context-hypothesis pairs using Gemini-assisted rewriting (Team et al., 2025), followed by strict human verification. The dataset is structured as a controlled minimal-pair experiment across four logical conditions, resulting in a total of N = 400 examples.

对于每个动词，我们使用 Gemini 辅助重写（Team 等，2025）生成上下文-假设对，随后进行严格的人工验证。该数据集构建为一个受控的最小对实验，涵盖四个逻辑条件，最终得到 N = 400 个示例。

To investigate whether teleological bias varies across different types of goal-directedness, we further annotated the 100 telic verbs into four fine-grained semantic classes (drawn on the theory of Levin, 1993): Change of State (44), Creation (39), Consumption (9), and Motion to Goal (8).

为了探讨目的论偏差是否在不同类型的目的性行为中有所不同，我们进一步将 100 个目的动词细分为四个语义类别（依据 Levin, 1993 的理论）：状态变化（44 个）、创造（39 个）、消耗（9 个）和向目标移动（8 个）。

We restricted this sub-classification to the telic group because, unlike homogeneous activities, telic predicates inherently encode a distinct resultant state (Beavers and Koontz-Garboden, 2012). This classification categorizes the nature of that result: Change of State verbs denote a transformation in the physical or abstract condition of an existing entity (e.g., melt, break); Creation verbs involve bringing an object into existence (e.g., build, knit); Consumption verbs entail the destruction or complete ingestion of an object (e.g., eat, burn) (Beavers, 2010); and Motion to Goal verbs describe movement towards a specific spatial endpoint (e.g., arrive, enter) (Talmy, 2000). This stratification enables us to probe whether LLMs are more prone to hallucinating completion for artifact-creating events, where the “teleological” purpose is most salient, compared to simple state changes.

我们之所以将这一子分类限制在目的性（telic）组，是因为与同质性活动不同，目的性谓语本质上编码了一个不同的结果状态（Beavers 和 Koontz-Garboden，2012）。这种分类描述了该结果的性质：状态变化动词表示现有实体的物理或抽象状态的转变（例如：融化、破碎）；创造动词涉及将一个物体带入存在（例如：建造、编织）；消耗动词意味着物体的毁灭或完全摄入（例如：吃、燃烧）（Beavers，2010）；而向目标运动动词描述的是向特定空间终点的移动（例如：到达、进入）（Talmy，2000）。这种分层结构使我们能够探讨，与简单的状态变化相比，语言模型是否更容易在创造物品的事件中产生幻觉，因为在这些事件中，“目的性” 目标最为显著。

Group	Condition	Premise Example	Hypothesis	Gold Label
A	Interrupted Accomp.(Telic + Cancel)	The carpenter was building a gazebo,but a storm destroyed the frame beforethe roof was on.	The carpenter built a gazebo.	Contradiction(□ False)
B	Interrupted Activity(Atelic + Stop)	The athletes were running on the track,but it started to hail.	The athletes ran on the track.	Entailment(□ True)
C	Ambiguous Accomp.(Telic + Process)	The carpenter was building a gazebo.	The carpenter built a gazebo.	Neutral(? Unknown)
D	Ambiguous Activity(Atelic + Process)	The athletes were running on the track.	The athletes ran on the track.	Entailment(□ True)

组	条件	前提示例	假设
A	中断的伴随（目的性 + 取消）	木匠正在建造一个凉亭，但一场风暴在屋顶安装之前就摧毁了框架。	木匠建了一个

B	中断的活动（非成就性 + 停止）	运动员们正在跑道上跑步，但开始下起了冰雹。	运动员们在跑
C	模糊的共犯（目的 + 过程）	木匠正在建造一个凉亭。	木匠建造了一
D	模糊活动（非事态性 + 过程）	运动员们在跑道上跑步。	运动员们在跑

Table 2: Examples from the IMPERFECTIVENLI dataset. Group C is the critical probe, where the imperfective paradox implies a Neutral relation, but teleological bias may lead models to predict Entailment.

表 2：来自 IMPERFECTIVENLI 数据集的例子。组 C 是关键探针，其中不完全体悖论暗示了中性关系，但目的论偏见可能导致模型预测蕴含关系。

Imperfective Logical Conditions. The core contribution of IMPERFECTIVENLI is its minimal-pair design, which can be viewed as a 2×2 matrix crossing Verb Telicity (Telic vs. Atelic) with Contextual Information (Interrupted vs. Ambiguous), described in the following groups A-D. Table 2 illustrates these conditions with examples.

未完成态逻辑条件。IMPERFECTIVENLI 的核心贡献在于其最小对设计，可以视为一个 2×2 矩阵，交叉 Verb Telicity（有界 vs. 无界）与 Contextual Information（被打断 vs. 不确定），如下面的 A-D 组所示。表 2 通过例子说明了这些条件。

Group A: Interrupted Accomplishment.

组 A：中断的成就。

This group serves as a counterfactual check. The premise describes a telic action in progress but includes an explicit cancellation clause (e.g., "...but a storm destroyed the frame before the roof was on."). This tests whether models can attend to explicit interruption of action and the negation of the result. The logical relation is Contradiction.

这个组用作反事实检验。前提描述了一个完成性动作正在进行，但包含了一个明确的取消条款（例如："……但暴风雨在屋顶安装之前摧毁了框架。"）。这测试模型是否能够关注到动作的明确中断以及结果的否定。逻辑关系是矛盾。

Group B: Interrupted Activity.

B 组：中断的活动。

This group controls for aspectual class properties. Due to the sub-interval property of atelic verbs, stopping an activity does not negate its occurrence (i.e., stopping running implies one has run). Even with an explicit interruption, the premise entails the hypothesis. The logical relation is Entailment.

这个组控制着方面类属性。由于非事态动词的子区间属性，停止一项活动并不否定其发生（即，停止跑步意味着已经跑过）。即使有明确的中断，前提也蕴含着假设。逻辑关系是蕴含。

Group C: Ambiguous Accomplishment.

C 组：模糊的成就。

This condition is the primary testbed for the Imperfective Paradox. Structurally, Group C is an ablation of Group A, where the interruption clause is removed, leaving only the progressive form of a telic verb. Semantically, the outcome is under-specified: the action was in progress, but completion is not known. The logical relation is Neutral. This group probes whether models hallucinate completion in the absence of explicit evidence.

这个条件是不完全体悖论的主要测试平台。从结构上看，Group C 是 Group A 的一个删减版本，去掉了中断从句，只保留了目的动词的进行体形式。从语义上看，结果是欠确定的：动作正在进行，但完成状态未知。逻辑关系为中性。该组探讨模型在缺乏明确证据的情况下是否会虚构完成状态。

Group D: Ambiguous Activity.

组 D：模糊活动。

Analogous to Group C, this condition is an ablation of Group B, retaining only the progressive form of an activity. Unlike Group C, the entailment holds here because the atelic process itself validates the event. This baseline tests whether models possess basic aspectual competence for atelic verbs. The logical relation is Entailment.

类似于 Group C，这一条件是对 Group B 的消减，仅保留了活动的进行体形式。与 Group C 不同，此处的蕴含关系成立是因为非事态过程本身验证了事件。这个基线测试模型是否具备对非事态动词的基本事态能力。逻辑关系为蕴含。

While standard NLI tasks utilize the labels Entailment, Contradiction, and Neutral (e.g., Bowman et al., 2015), we map these to the natural

虽然标准的自然语言推理（NLI）任务使用 Entailment（蕴含）、Contradiction（矛盾）和 Neutral（中性）这些标签（例如，Bowman 等，2015），但我们将其映射到自然

language truth values “True”, “False”, and “Unknown” respectively for our experiments.

在我们的实验中，分别使用 “True”、“False” 和 “Unknown” 来表示逻辑真值。

Quality Assurance. To ensure the validity and naturalness of the generated data, we recruited three native English speakers as evaluators via Prolific [2]. They assessed the sentences across three linguistic dimensions, Grammar, Fluency, and Adequacy (details in Appendix C), and were compensated at a rate of £9/hour. Beyond quantitative ratings, evaluators provided corrections for any unnatural or unclear instances. Discrepancies and conflicting revisions were resolved through manual verification, ensuring quality in the final dataset.

质量保证。为了确保生成数据的有效性和自然性，我们通过 Prolific [2] 雇请了三位母语为英语的评估者。他们从语法、流畅性和充分性三个语言维度对句子进行评估（详情见附录 C），并以每小时 9 英镑的费率获得报酬。除了定量评分外，评估者还会对任何不自然或不清晰的实例进行更正。对于存在分歧或冲突的修改，我们通过人工验证来解决，从而确保最终数据集的质量。

4 Experimental Settings

4 实验设置

4.1 Models

4.1 模型

We experimented with 7 instruction-tuned open-weight models with their parameter around 7-9B: Llama-3.1-8B-Instruct (Grattafiori et al., 2024), Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), Qwen2.5-7B-Instruct (Qwen et al., 2025), deepseek-1lm-7b-chat (DeepSeek-AI, 2024), gemma-2-9b-it (Team et al., 2024), glm-4-9b-chat-hf (GLM et al., 2024), Yi-1.5-9B-Chat (AI et al., 2025).

我们尝试了 7 个指令调优的开放权重模型，其参数量约为 7-9B：Llama-3.1-8B-Instruct（Grattafiori 等，2024），Mistral-7B-Instruct-v0.3（Jiang 等，2023），Qwen2.5-7B-Instruct（Qwen 等，2025），deepseek-1lm-7b-chat（DeepSeek-AI，2024），gemma-2-9b-it（Team 等，2024），glm-4-9b-chat-hf（GLM 等，2024），Yi-1.5-9B-Chat（AI 等，2025）。

4.2 Prompts

4.2 提示词

To disentangle the model's intrinsic biases from its instruction-following capabilities, we evaluate four distinct prompting strategies. The detailed prompts given to the models are presented in Appendix B.

为了将模型的内在偏见与其遵循指令的能力区分开来，我们评估了四种不同的提示策略。提供给模型的详细提示见附录 B。

(1) Strict Logic Prompt (Zero-Shot). Our primary setting employs a simple zero-shot instruction, framing the model as a "strict logician". This prompt explicitly prohibits common sense assumptions (e.g., the pragmatic inference that "building

(1) 严格逻辑提示（零样本）。我们的主要设置采用了一个简单的零样本指令，将模型框定为一个 "严格逻辑学家"。这个提示明确禁止使用常识性假设（例如，"建造" 这一实用推论）。

usually implies a house") and forces the model to rely solely on the provided text.

通常意味着一座房子")，并迫使模型仅依赖所提供的文本。

(2) Definition-Aware Prompt (DAP). To test whether the failure stems from a lack of linguistic knowledge or a failure of application, we explicitly inject the linguistic definition of the imperfective aspect for both Activity and Accomplishment into the system prompt. If the bias persists even when the rule is provided, it suggests that the teleological prior is robust enough to override in-context instructions.

(2) 基于定义的提示（DAP）。为了测试失败是由于缺乏语言知识还是应用失败，我们将不完美体的语义定义明确注入到系统提示中，用于活动和成就。如果在提供规则后偏差仍然存在，这表明目的性先验足够强大，可以覆盖上下文中的指令。

(3) Chain-of-Thought Prompt (CoT). We adapt the standard "Let's think step by step" paradigm (Wei et al., 2022b). This setting investigates whether the Teleological Bias is a result of fast, superficial processing. We hypothesize that by forcing the model to articulate the temporal status of the event before predicting the label, it may self-correct the hallucination for Ambiguous Accomplishment.

(3) 思维链提示（CoT）。我们采用标准的 "逐步思考" 范式（Wei 等，2022b）。这种设置探讨了目的论偏差是否是由于快速、表面的处理所致。我们假设，通过迫使模型在预测标签之前阐述事件的时间状态，可能会自行纠正 "模糊成就" 的幻觉。

(4) Counterfactual Prompt (Counterfactual). Drawing on Dowty (1977)'s theory that the progressive aspect presupposes an "inertia world" (where events proceed normally), this prompt attempts to explicitly break this assumption. We explicitly instruct the model to first generate three plausible real-world scenarios in which the event is interrupted and its endpoint is not reached, before making a judgment.

(4) 反事实提示（反事实）。借鉴 Dowty（1977）的理论，即进行体（progressive aspect）预设了一个 "惯

性世界”（其中事件按正常方式发展），该提示试图明确地打破这一假设。我们明确地指示模型首先生成三个合理的现实场景，在这些场景中事件被中断且未达到终点，然后再进行判断。

4.3 Metrics

4.3 指标

We report following metrics:

我们报告以下指标：

(1) Group-wise Accuracy (ACC): We calculate classification accuracy separately for each logical condition (Groups A–D).
(2) Teleological Bias Rate (TBR): This metric specifically quantifies the “Hallucination of Completion” in Group C (Ambiguous Accomplishment). Since the Gold Label for Group C is Unknown, any prediction of True indicates that the model has hallucinated a result. We define TBR as:

(1) 分组准确率（ACC）：我们分别对每个逻辑条件（组 A 至 D）计算分类准确率。
(2) 目的性偏差率（TBR）：该指标专门量化组 C（模糊完成）中的 “完成幻觉”。由于组 C 的黄金标签为未知，任何预测为真的结果都表明模型产生了幻觉结果。我们定义 TBR 为：

$$\mathrm{TBR}_C = \frac{\sum_{i \in C} \mathbb{I}(\hat{y}_i = \mathrm{True})}{|C|} \tag{1}$$

where C is the set of examples in Group C, and $\hat{y}_i$ is the model’s prediction. A higher TBR indicates a stronger reliance on teleological priors.

其中 C 是 Group C 中的例子集合，而 $\hat{y}_i$ 是模型的预测结果。较高的 TBR 表明对目的论先验的依赖更强。

(3) Aspectual Awareness Gap (Δ_{AA}): This metric measures the model’s capability to distinguish

(3) 时态意识差距 (Δ_{AA})：该指标衡量模型区分能力

the logical implications of atelic processes (which imply realization) from telic processes (which do not). It is calculated as the difference between the accuracy on the Ambiguous Activity baseline (Group D) and the error rate on the Ambiguous Accomplishment probe (Group C):

非谓体过程（其隐含实现）的逻辑含义与谓体过程（其不隐含实现）的逻辑含义。其计算方式为：模糊活动基线（组 D）上的准确率与模糊完成度探针（组 C）上的错误率之间的差异：

$$\Delta_{AA} = \mathrm{ACC}_D - \mathrm{TBR}_C \tag{2}$$

where ACC_D represents the correct entailment rate for Activity verbs. A high gap indicates ideal aspectual reasoning: the model correctly infers entailment for activities while correctly suspending judgment (low bias) for accomplishments. A low gap suggests that the model treats all progressive verbs homogeneously, likely applying a shallow heuristic (e.g., “mentioning an action implies its completion”) regardless of lexical aspect.

其中 ACC_D 表示对活动动词的正确蕴含率。较大的差距表明理想的方面推理：模型在处理活动时正确推断出蕴含关系，而在处理成就时正确保持谨慎（低偏差）。较小的差距则表明模型将所有进行体动词视为同质的，可能应用了浅层启发式规则（例如，“提及一个动作即意味着其完成”），而不论词法方面如何。

Note that Δ_{AA} is most interpretable when ACC_D is high; a low Δ_{AA} driven by a low ACC_D rather than a high TBR_C reflects a distinct failure mode, namely, an inability to recognize valid entailments for atelic verbs, rather than teleological overgeneralization.

请注意，当 ACC_D 较高时，Δ_{AA} 最具可解释性；由较低的 ACC_D 导致的较低 Δ_{AA}，而非由较高的 TBR_C 所驱动，反映了一种不同的故障模式，即无法识别非事态动词的有效蕴含，而不是目的论的过度泛化。

5 Results and Analysis

5 结果与分析

In this section, we present the evaluation results on IMPERFECTIVENLI. We aim to answer four key questions: whether LLMs exhibit teleological bias, how prompting strategies mitigate this bias, how the model scale affects the bias, and how these biases manifest across different semantic categories.

在本节中，我们展示了在 IMPERFECTIVENLI 上的评估结果。我们旨在回答四个关键问题：语言模型是否表现出目的论偏差，提示策略如何缓解这种偏差，模型规模如何影响这种偏差，以及这些偏差在不同语义类别中的表现如何。

5.1 The Dominance of Teleological Bias for “If A Subject was V-ing, then the Subject V-ed” in Zero-shot

5.1 零样本任务中 “如果主体正在 V，那么主体 V 了” 的目的论偏见占主导地位

The first few rows of Table 3 present the performance of seven open-weight models in the zero-shot setting. The results reveal a fundamental inability of current LLMs to decouple the description of a process from the assertion of its result.

表 3 的前几行展示了七种开放权重模型在零样本设置下的性能。结果揭示了当前 LLMs 在将过程的描述与其结果的断言分离方面存在根本性的不足。

$P(\text{Complete} \mid \text{Past Progressive}) \approx 1? - -$The Illusion of Competence. At first glance, models appear competent in aspectual reasoning, achieving near-perfect accuracy on Group D for ambiguous Activity (atelic verbs) (e.g., Llama-3.1: 0.98, Mistral: 1.00). However, this high performance is deceptive. When contrasted with the catastrophic failure on Group C for ambiguous Accomplishment (telic verbs) (e.g., Llama-3.1: 0.02), a different picture emerges. The near-zero Δ_{AA} for most

$P(\text{Complete} \mid \text{Past Progressive}) \approx 1? - -$The Illusion of Competence. 乍一看，模型在方面推理方面似乎表现得相当出色，在模糊活动（非谓语动词）（例如，Llama-3.1：0.98，Mistral：1.00）的 Group D 中实现了接近完美的准确率。然而，这种高性能是具有欺骗性的。当与在模糊成就（目的性动词）（例如，Llama-3.1：0.02）的 Group C 中出现灾难性失败进行对比时，呈现出不同的图景。对于大多数模型来说，接近零的 Δ_{AA}

Prompt	Model	Acc A (□)	Acc B (□)	Acc C (□)	Acc D (□)	TBR $_C$ (□)	Δ_{A}
Zero-shot	Llama-3.1-8B-Instruct	0.47	0.85	0.02	0.98	0.98	0.0
Mistral-7B-Instruct-v0.3	0.37	0.92	0.02	1.00	0.97	0.03	
Qwen2.5-7B-Instruct	0.20	0.98	0.47	0.97	0.53	0.44	
Yi-1.5-9B-Chat	0.35	0.94	0.02	1.00	0.98	0.02	
deepseek-llm-7b-chat	0.04	0.88	0.00	1.00	1.00	0.00	

gemma-2-9b-it	0.03	0.96	0.06	1.00	0.94	0.06	
glm-4-9b-chat-hf	0.14	0.98	0.03	1.00	0.97	0.03	
DAP	Llama-3.1-8B-Instruct	0.60	0.95	0.36	0.99	0.45	0.54
Mistral-7B-Instruct-v0.3	0.55	0.96	0.06	1.00	0.94	0.06	
Qwen2.5-7B-Instruct	0.35	0.97	0.89	0.72	0.09	0.63	
Yi-1.5-9B-Chat	0.57	0.97	0.20	1.00	0.80	0.20	
deepseek-llm-7b-chat	0.12	0.92	0.00	1.00	1.00	0.00	
gemma-2-9b-it	0.08	0.90	0.60	0.88	0.40	0.48	
glm-4-9b-chat-hf	0.08	1.00	0.33	0.98	0.67	0.31	
CoT	Llama-3.1-8B-Instruct	0.15	0.39	0.67	0.65	0.33	0.32
Mistral-7B-Instruct-v0.3	0.23	0.65	0.41	0.66	0.56	0.10	
Qwen2.5-7B-Instruct	0.30	0.80	0.97	0.37	0.03	0.34	
Yi-1.5-9B-Chat	0.56	0.81	0.85	0.44	0.13	0.31	
deepseek-llm-7b-chat	0.05	0.88	0.40	0.94	0.60	0.34	
gemma-2-9b-it	0.15	0.88	0.98	0.15	0.02	0.13	
glm-4-9b-chat-hf	0.24	0.95	0.38	0.97	0.62	0.35	
Counterfactual	Llama-3.1-8B-Instruct	0.35	0.00	0.97	0.00	0.00	0.00
Mistral-7B-Instruct-v0.3	0.28	0.39	0.87	0.19	0.11	0.08	
Qwen2.5-7B-Instruct	0.07	0.57	1.00	0.02	0.00	0.02	
Yi-1.5-9B-Chat	0.07	0.41	1.00	0.10	0.00	0.10	
deepseek-llm-7b-chat	0.25	0.97	0.00	0.98	1.00	-0.02	
gemma-2-9b-it	0.03	0.00	1.00	0.00	0.00	0.00	
glm-4-9b-chat-hf	0.23	0.66	0.73	0.24	0.20	0.04	

提示	模型	A 级（□）	Acc B (□)	ACC C (□)	ACC D (□)	TBR $_C$ (□)	Δ
零样本	Llama-3.1-8B-Instruct	0.47	0.85	0.02	0.98	0.98	0.0
Mistral-7B-Instruct-v0.3	0.37	0.92	0.02	1.00	0.97	0.03	
Qwen2.5-7B-Instruct	0.20	0.98	0.47	0.97	0.53	0.44	
Yi-1.5-9B-Chat	0.35	0.94	0.02	1.00	0.98	0.02	
deepseek-llm-7b-chat	0.04	0.88	0.00	1.00	1.00	0.00	
gemma-2-9b-it	0.03	0.96	0.06	1.00	0.94	0.06	
glm-4-9b-chat-hf	0.14	0.98	0.03	1.00	0.97	0.03	
DAP	Llama-3.1-8B-Instruct	0.60	0.95	0.36	0.99	0.45	0.5
Mistral-7B-Instruct-v0.3	0.55	0.96	0.06	1.00	0.94	0.06	
Qwen2.5-7B-Instruct	0.35	0.97	0.89	0.72	0.09	0.63	
Yi-1.5-9B-Chat	0.57	0.97	0.20	1.00	0.80	0.20	
deepseek-llm-7b-chat	0.12	0.92	0.00	1.00	1.00	0.00	
gemma-2-9b-it	0.08	0.90	0.60	0.88	0.40	0.48	
glm-4-9b-chat-hf	0.08	1.00	0.33	0.98	0.67	0.31	
思考链（CoT）	Llama-3.1-8B-Instruct	0.15	0.39	0.67	0.65	0.33	0.3

Mistral-7B-Instruct-v0.3	0.23	0.65	0.41	0.66	0.56	0.10	
Qwen2.5-7B-Instruct	0.30	0.80	0.97	0.37	0.03	0.34	
Yi-1.5-9B-Chat	0.56	0.81	0.85	0.44	0.13	0.31	
deepseek-llm-7b-chat	0.05	0.88	0.40	0.94	0.60	0.34	
gemma-2-9b-it	0.15	0.88	0.98	0.15	0.02	0.13	
glm-4-9b-chat-hf	0.24	0.95	0.38	0.97	0.62	0.35	
反事实	Llama-3.1-8B-Instruct	0.35	0.00	0.97	0.00	0.00	0.0
Mistral-7B-Instruct-v0.3	0.28	0.39	0.87	0.19	0.11	0.08	
Qwen2.5-7B-Instruct	0.07	0.57	1.00	0.02	0.00	0.02	
Yi-1.5-9B-Chat	0.07	0.41	1.00	0.10	0.00	0.10	
deepseek-llm-7b-chat	0.25	0.97	0.00	0.98	1.00	-0.02	
gemma-2-9b-it	0.03	0.00	1.00	0.00	0.00	0.00	
glm-4-9b-chat-hf	0.23	0.66	0.73	0.24	0.20	0.04	

Table 3: Model performance under Zero-shot, CoT, DAP, and Counterfactual prompting strategies with Acc for A-D Groups, Teleological Bias Rate (TBR), and Aspectual Awareness Gap (Δ_{AA}).

表 3：在零样本、推理链、DAP 和反事实提示策略下，A-D 组的准确率（Acc）、目的性偏倚率（TBR）和方面意识差距（Δ_{AA} ）。

models indicates that they do not genuinely understand the Sub-interval Property of the atelic verbs (Bennett and Partee, 1978). Instead, they likely employ a shallow heuristic: "If Subject was V-ing, then Subject V-ed", regardless of verbs in the Zero-shot setting. This aligns with findings by the NLI work McCoy et al. (2019), suggesting that the models might often rely on lexical overlap strategies rather than deep semantic compositionality.

models 表明它们并未真正理解非谓语动词的子区间性质（Bennett 和 Partee，1978）。相反，它们可能采用了一种浅层的启发式方法："如果主语正在 V-ing，那么主语 V-ed"，而不管零样本设置中的动词。这与 NLI 工作 McCoy 等人（2019）的发现一致，表明模型可能经常依赖于词汇重叠策略，而不是深层次的语义组合性。

Contextual Neglect via Semantic Priming. Perhaps the most alarming finding is the resistance to explicit negation in Group A (Interrupted Accomplishment). Despite the textually grounded interruption (e.g., "…but a storm destroyed it"), most models exhibit a stark performance degradation. This stands in sharp contrast to Group B (Interrupted Activity), where models maintain high accuracy. This performance gap is semantically revealing: for activities, an interruption does not negate the event's occurrence (the sub-interval property ensures that "stopped running" entails "ran"), whereas for accomplishments, interruption strictly precludes the result.

语义启动下的情境忽视。也许最令人不安的发现是，Group A（被打断的成就）对显性否定表现出的抗拒。尽管文本中明确指出了中断事件（例如，"……但一场风暴摧毁了它"），但大多数模型的表现却显著下降。这与 Group B（被打断的活动）中模型保持高准确率形成了鲜明对比。这种性能差异在语义上具有揭示性：对于活动而言，中断并不否定事件的发生（子区间属性确保了 "停止跑步" 意味着 "跑步"），而对于成就而言，中断则严格排除了结果的发生。

The failure in Group A suggests that the models' reasoning is driven by Semantic Priming rather

Group A 的失败表明，模型的推理过程受到语义启动（Semantic Priming）的驱动，而不是

than compositional logic (Ettinger, 2020; Kassner and Schütze, 2020). We hypothesize that the co-occurrence of an agent (e.g., carpenter) and a telic verb (e.g., building) activates the "completed result" representation in the model's high-dimensional space with such magnitude that it overrides the attention mechanism's focus on the subsequent cancellation clause. This phenomenon exemplifies a failure of contextual fidelity, where static parametric knowledge priors overpower the dynamic input context (Mallen et al., 2023).

相比于构成逻辑（Ettinger, 2020; Kassner 和 Schütze, 2020）。我们假设，一个主体（例如木匠）和一个目的动词（例如建造）的共现，会在模型的高维空间中激活 “完成结果” 的表征，其强度足以覆盖注意力机制对后续取消从句的关注。这一现象体现了上下文忠实性的失败，其中静态的参数化知识先验超越了动态输入上下文（Mallen 等，2023）。

The Outlier: Qwen2.5 shows exceptional performance. Qwen2.5 stands as a notable exception, maintaining a balanced performance with a TBR of 0.53 and a healthy Δ_{AA} of 0.44. Unlike its peers, Qwen demonstrates an emerging ability to suspend judgment in ambiguous contexts.

异常值：Qwen2.5 表现出色。Qwen2.5 是一个显著的例外，保持了平衡的表现，其 TBR 为 0.53，健康度为 0.44。与同辈不同，Qwen 展现出在模糊情境中暂停判断的新兴能力。

5.2 Trade-off: Bias Mitigation vs. Semantic Calibration in Advanced Prompts

5.2 权衡：在高级提示中的偏见缓解与语义校准

Evaluating three advanced prompting strategies reveals a consistent trade-off between mitigating teleological bias and maintaining semantic calibration across aspectual classes.

评估三种先进的提示策略表明，在不同方面类别间减轻目的论偏差和保持语义校准之间存在一致的权衡。

DAP: The Knowledge-Application Gap. The DAP which introduces expert linguistic definition for the aspectual classes yields mixed results, exposing a critical gap between possessing linguistic knowledge and applying it. While supplying the linguistic rule improves Llama-3.1's accuracy on Group C (0.02 □ 0.36), it remains ineffective for Mistral (0.06). This persistence of bias implies that the teleological prior is not a misunderstanding of the task definition but a robust feature of the model's probability distribution. As noted by Dziri et al. (2023), compositional reasoning in Transformers often degrades when it conflicts with high-frequency statistical patterns (i.e., the reporting bias) seen during pre-training.

DAP：知识与应用之间的差距。引入专家语言学定义的 DAP 在方面类别的应用中产生了混合的结果，暴露出在拥有语言学知识和实际应用之间存在一个关键的差距。虽然提供语言规则提高了 Llama-3.1 在组 C 上的准确性（0.02 □ 0.36），但对于 Mistral 却仍然无效（0.06）。这种偏见的持续存在表明，目的性先验并不是对任务定义的误解，而是模型概率分布的一个稳健特征。正如 Dziri 等人（2023）所指出的，在 Transformer 中，当与预训练期间观察到的高频统计模式（即报告偏差）发生冲突时，组合推理往往会退化。

CoT: Reasoning-Induced Uncertainty. CoT prompting successfully reduces the TBR for most models, proving that allocating compute to intermediate reasoning helps suppress System-1 heuristics (Wei et al., 2022b). However, this comes at a cost. We observe a significant degradation in Group D (Activities), where accuracy drops substantially (e.g., Llama-3.1 0.98 □ 0.65). Qualitative analysis reveals a phenomenon of "over-reasoning" or unfaithful reasoning (Turpin et al., 2023). When forced to analyze "step-by-step", models often hallucinate hypothetical interruptions for simple activities. [3] CoT appears to over-generalize the progressive-as-incomplete heuristic: when forced to reason step-by-step, models often impose hypothetical interruptions even on atelic verbs, inadvertently suppressing valid entailments.

CoT：推理诱导的不确定性。CoT 提示成功地降低了大多数模型的 TBR，证明了将计算资源分配给中间推理有助于抑制系统 1 的启发式方法（Wei 等，2022b）。然而，这需要付出代价。我们观察到在 Group D（活动）中出现了显著的性能下降，其中准确率大幅下降（例如，Llama-3.1 0.98 □ 0.65）。定性分析揭示了一种 “过度推理” 或不忠实推理的现象（Turpin 等，2023）。当模型被强制要求进行 “逐步分析” 时，它

们经常对简单的活动虚构假设的中断。[3] CoT 似乎过度泛化了 “逐步即未完成” 的启发式方法：当模型被强制进行逐步推理时，它们经常甚至对非持续性动词施加假设的中断，无意中抑制了有效的蕴含关系。

Counterfactuals: The Calibration Crisis. The Counterfactual prompt represents the most aggressive intervention for arbitrary interruption. While highly effective at curing teleological bias (Group C Acc □ 1.00), it induces an obvious calibration bias. Accuracy on Group D collapses to near zero for Llama-3.1. We show the example of Llama-3.1 in Figure 1. The models swing from being “naively optimistic” (Zero-shot) to “paranoidly skeptical” (Counterfactual). The Counterfactual clearly over-instructed the models to follow the imperfective patterns of the telic verbs in Group C, resulting in misunderstanding for the atelic verbs in Group D. The models failed to dynamically adjust their reasoning strategy based on

反事实：校准危机。反事实提示代表了对任意中断最激进的干预。虽然在消除目的论偏差（组 C 准确率 □1.00）方面非常有效，但它会导致明显的校准偏差。对于 Llama-3.1 来说，组 D 的准确率几乎降为零。我们在图 1 中展示了 Llama-3.1 的例子。模型从 “天真乐观”（零样本）转变为 “极端怀疑”（反事实）。反事实明显让模型过度遵循了组 C 中动词的非完成体模式，导致对组 D 中非目的动词的理解出现偏差。模型未能根据

the (a)telicity of verbs, resulting in a wholesale rejection of progressive entailments. This highlights a lack of (aspectual) calibration (Kadavath et al., 2022; Kapoor et al., 2024): while the counterfactual prompt successfully suppresses teleological bias for telic verbs, models fail to modulate this skepticism based on verb class, applying it indiscriminately to atelic verbs as well. Detailed analyses and visualization for other models are provided in Appendix E.

动词的非趋向性（atelicity），导致对进行体蕴含（progressive entailments）的全面拒绝。这突显了在（时态方面）校准上的不足（Kadavath 等，2022；Kapoor 等，2024）：虽然反事实提示成功地抑制了趋向性动词的目的论偏差（teleological bias），但模型未能根据动词类别调整这种怀疑态度，而是无差别地将其应用于非趋向性动词。其他模型的详细分析和可视化结果见附录 E。

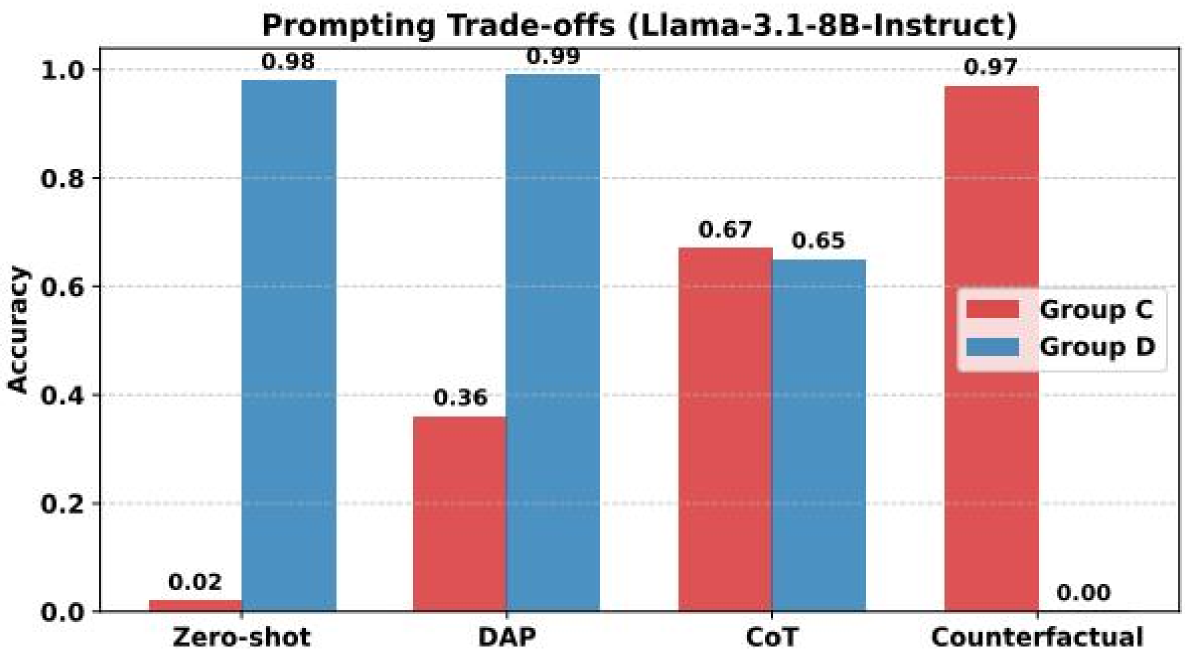

Figure 1: images/0128822c8352a5a71e028b6e7f2ad2b4f2ad91cd405f2b4af6b336aeb2845c13.jpg

Figure 1: The trade-off in prompting (Llama-3.1). As prompts become more aggressive in targeting bias (Zero-shot □ Counterfactual), accuracy on Group C improves significantly (Pearson r = 1.00, p = 0.00), while accuracy on Group D declines, though less consistently (r = -0.91, p = 0.09).

This trade-off illustrates the calibration bias.

图 1：提示中的权衡（Llama-3.1）。随着提示在针对偏见方面变得更加激进（从零样本到反事实），对 Group C 的准确性显著提高（皮尔逊 $r = 1.00$，$p = 0.00$），而对 Group D 的准确性则下降，但不如 Group C 一致（$r = -0.91$，$p = 0.09$）。这种权衡体现了校准偏差。

Summary on Different Prompts. In summary, current LLMs struggle to balance the Imperfective Paradox. They oscillate between two extremes: naive teleology (Zero-shot) where everything is completed, and paranoid skepticism (Counterfactual) where nothing is ever certain (see the trends in Figure 1). This inability to decouple telicity from entailment highlights a critical gap in the true aspectual reasoning capabilities of LLMs.

不同提示的总结。总而言之，当前的 LLMs 在平衡非完满悖论方面存在困难。它们在两个极端之间摇摆：天真目的论（零次提示）中一切皆已完成，以及偏执怀疑论（反事实提示）中一切皆不确定（参见图 1 中的趋势）。这种无法将目的性与蕴含性分离的能力，凸显了 LLMs 在真实方面推理能力方面的关键缺陷。

5.3 Impact of Model Scale

5.3 模型规模的影响

To understand whether aspectual reasoning is a capability that improves linearly with scale or emerges as a distinct skill past a certain threshold, we evaluate the Qwen2.5 family, which showed better performance in §5.1, across five distinct parameter sizes: 1.5B, 7B, 14B, 32B, and 72B. The results, detailed in Table 5 (Appendix) and visualized in Figure 2, reveal that aspectual reasoning follows a distinct trajectory of emergence.

为了理解方面推理能力是否随着规模的增加而线性提升，还是在达到某个阈值后作为一种独立的技能出现，我们评估了在第 5.1 节中表现出更好性能的 Qwen2.5 系列模型，该模型在五个不同的参数规模下进行了测试：1.5B、7B、14B、32B 和 72B。结果详见附录中的表 5，并在图 2 中进行了可视化，揭示了方面推理能力呈现出一种独特的出现轨迹。

We observe a clear monotonic relationship between model scale and both evaluation metrics. Model size is strongly negatively correlated with Teleological Bias Rate (TBR; $r = -0.93$, $p =$

我们观察到模型规模与两个评估指标之间存在明显的单调关系。模型规模与目的论偏差率（TBR；$r = -0.93$，$p =$

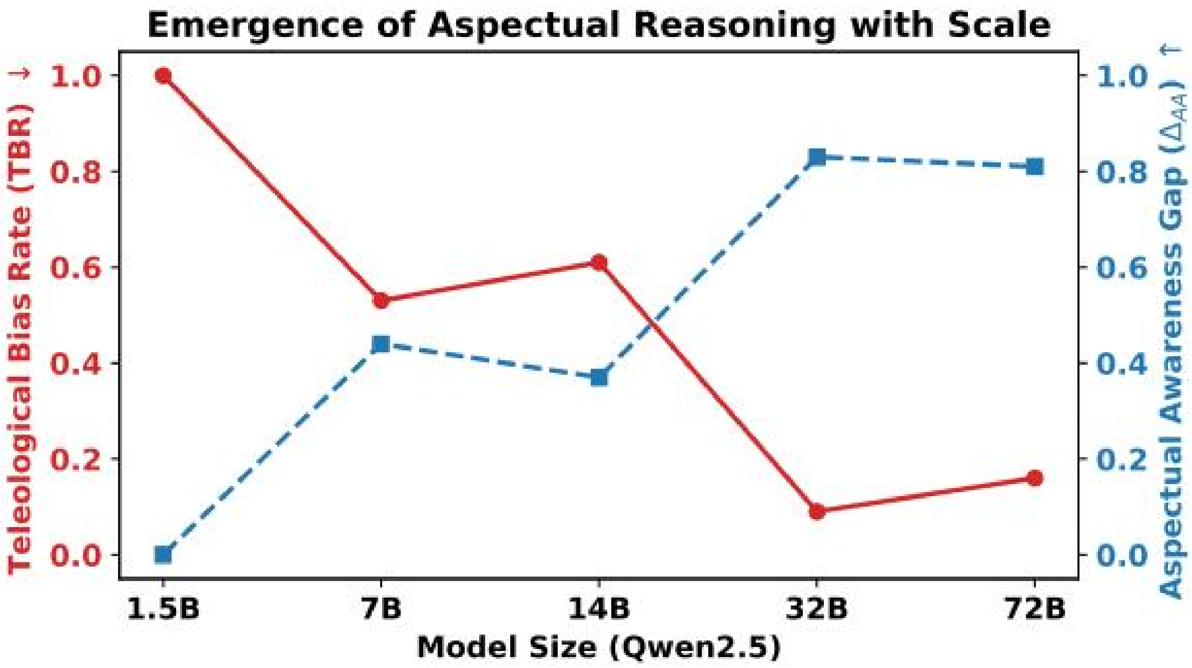

Figure 2: images/ddb6d50b97379510debe39843a8a42421e41e6904eef4386c1362f8346829a88.jpg

Figure 2: The emergence of aspectual reasoning capabilities across model scales. It shows a significant decrease in TBR and a corresponding increase in Δ_{AA} .

图 2：模型规模下方面推理能力的出现。它显示了 TBR 的显著下降以及相应的 Δ_{AA} 增加。

0.02) and strongly positively correlated with the Aspectual Awareness Gap (Δ_{AA} ; r = 0.94, p = 0.01), indicating that increased model capacity systematically reduces teleological overgeneralization while enhancing aspect-sensitive reasoning.

0.02) 且与方面意识差距 (Δ_{AA} ; r = 0.94, p = 0.01) 显著正相关，表明模型容量的增加系统性地减少了目的论过度泛化，同时增强了方面敏感性推理能力。

The 1.5B model exhibits a complete inability to handle the Imperfective Paradox. With a TBR of 1.00 and a Δ_{AA} of 0.00, it treats every progressive verb as an entailed completion. As model scale increases, a dramatic "phase shift" (Wei et al., 2022a) is observed, especially at the 32B level: Accuracy on Group C surges to 0.91, and the Δ_{AA} peaks at 0.83. This suggests that robust aspectual distinction, the ability to selectively inhibit the entailment for telic verbs while maintaining it for atelic ones, is an emergent ability. Although the 72B model maintains strong performance, Δ_{AA} decreases slightly to 0.81, suggesting a potential saturation or optimization bottleneck rather than further qualitative gains. Overall, these results indicate that aspectual understanding improves with scale and exhibits non-linear emergence characteristics rather than linear scaling.

1.5B 模型表现出完全无法处理未完成体悖论的缺陷。其 TBR 为 1.00，且 Δ_{AA} 为 0.00，它将每个进行体动词都视为蕴含完成。随着模型规模的增加，观察到一个显著的 "相位转变"（Wei 等，2022a），尤其是在 32B 级别时：Group C 的准确率飙升至 0.91，而 Δ_{AA} 达到 0.83 的峰值。这表明，对时态的稳健区分能力，即能够选择性地抑制终止性动词的蕴含，同时保留非终止性动词的蕴含，是一种涌现能力。虽然 72B 模型保持了强劲的表现，但 Δ_{AA} 略微下降至 0.81，这表明可能存在饱和或优化瓶颈，而不是进一步的质性提升。总体而言，这些结果表明，时态理解能力随着规模的增加而提升，并表现出非线性的涌现特性，而非线性扩展。

5.4 Fine-grained Analysis on Verb Semantics

5.4 对动词语义的细粒度分析

Do all telic verb classes trigger the same degree of teleological bias? To answer this, we disaggregate the performance on Group A and Group C based on the four semantic sub-classes defined in §3: Creation, Consumption, Change of State, and Motion to Goal. Figure 3 visualizes the average accuracy across all evaluated models in the zero-shot setting. Detailed results for each model as well as for other prompt settings can be found in Appendix E.

所有目的性动词类是否都会引发相同程度的目的论偏见？为回答这个问题，我们根据第 3 节中定义的四个语义子类（创造、消费、状态变化和向目标运动）对 Group A 和 Group C 的表现进行分解。图 3 展示了零样本设置下所有评估模型的平均准确率。每个模型的详细结果以及其他提示设置的结果可在附录 E 中找到。

The Resilience of Motion vs. The Creation Penalty. We observe a significant semantic divergence. Verbs of Motion to Goal (e.g., “arrive”, “en-

运动的韧性与创造的代价。我们观察到显著的语义差异。趋向目标的动词（例如，“到达”、“进入”）

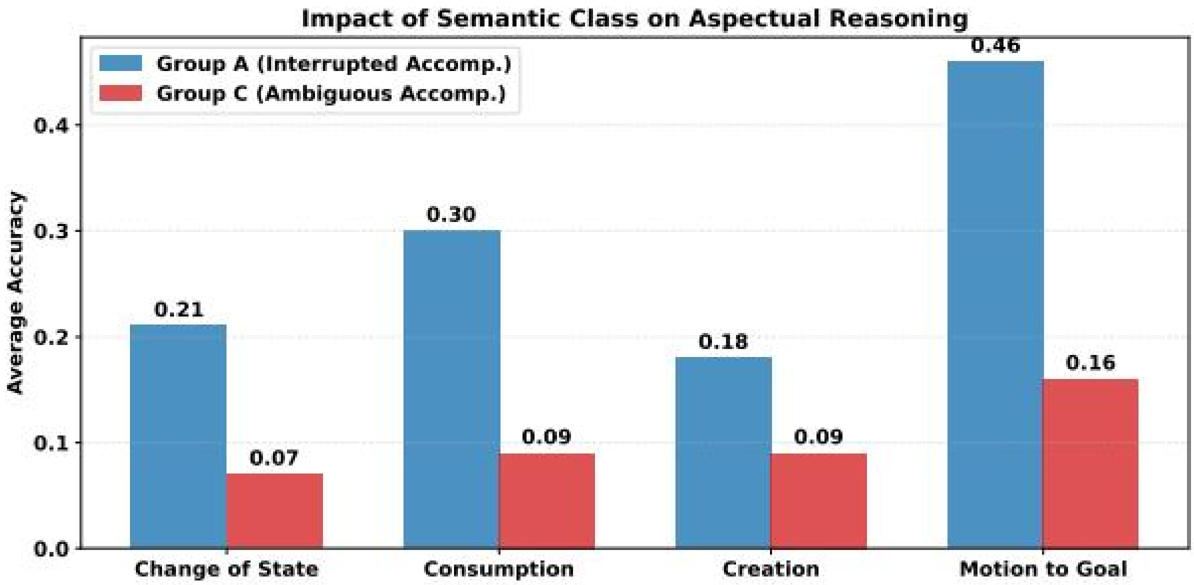

Figure 3: images/e573b4eaf2d17ef36aa6f5b4ab0c150aa21e8d28b8e33e73c8240edad46ef0c3.jpg

Figure 3: Average accuracy across models by semantic verb classes in Zero-shot. Motion verbs are consistently easier for models to reason about, while Creation verbs induce strong teleological bias (low accuracy in Group C) and resistance to contextual cancellation (lowest accuracy in Group A).

图 3：零样本条件下，按语义动词类别划分的模型平均准确率。运动动词对于模型来说始终更容易推理，而创造动词则会引发强烈的目的论偏差（在组 C 中准确率较低）并表现出对上下文抵消的抗拒（在组 A 中准确率最低）。

ter”) consistently yield the highest performance. In Group A, the average accuracy for Motion to Goal verbs is 46%, significantly outperforming Creation (18%) and Change of State (21%). The performance gap is substantial: $\Delta_{\text{Motion-Creation}} \approx +28\%$ in Group A and $\approx +13\%$ in Group C. [4]

ter”）一直表现出最佳性能。在 Group A 中，Motion to Goal 动词的平均准确率为 46%，显著优于 Creation

(18%) 和 Change of State (21%)。性能差距非常明显：Group A 中为 $\Delta_{\text{Motion-Creation}} \approx +28\%$，Group C 中为 $\approx +13\%$。[4]

We hypothesize that this stems from the nature of the resultative state. For Creation verbs (e.g., “build”), the result involves the existential realization of an object. This explicit “coming into being” appears to act as a stronger attractor in the language model’s predictive distribution than a mere change in spatial coordinates involved in Motion. Consequently, models exhibit higher resistance to negation when an artifact’s existence is at stake.

我们假设这是因为结果状态的本质。对于表示创造的动词（例如 “建造”），结果涉及一个物体的存在实现。这种明确的 “产生” 似乎在语言模型的预测分布中比运动中涉及的空间坐标变化更具吸引力。因此，当物品的存在受到质疑时，模型在否定方面表现出更高的抵抗性。

Representational Analysis: Encoding vs. Inference. Does teleological bias reflect a failure to distinguish process from result in the model’s internal representations, or does it arise later, at the point of inference? In order to probe this, we measured the cosine similarity between contextual embeddings of progressive (e.g., “was building”) and perfective (e.g., “built”) verb phrases using bert-base-uncased (Devlin et al., 2019), treating representational similarity as a proxy for how well the model’s encoder separates the two aspectual forms.

表征分析：编码与推理。这种目的论偏差是否反映了模型内部表征中未能区分过程与结果，还是在推理阶段才出现？为了探讨这个问题，我们使用 bert-base-uncased (Devlin 等人，2019) 测量了渐进式（例如，“was building”）和完成式（例如，“built”）动词短语的上下文嵌入之间的余弦相似度，将表征相似度作为模型编码器区分这两种时态形式效果的代理指标。

Figure 4 reveals a striking inverse relationship (Pearson r = -0.97, p = 0.03) between how similarly a model encodes the progressive and perfec-

图 4 揭示了一个显著的反向关系（Pearson r = -0.97, p = 0.03），即模型在编码渐进性和完美性方面越相似，其性能表现越差。

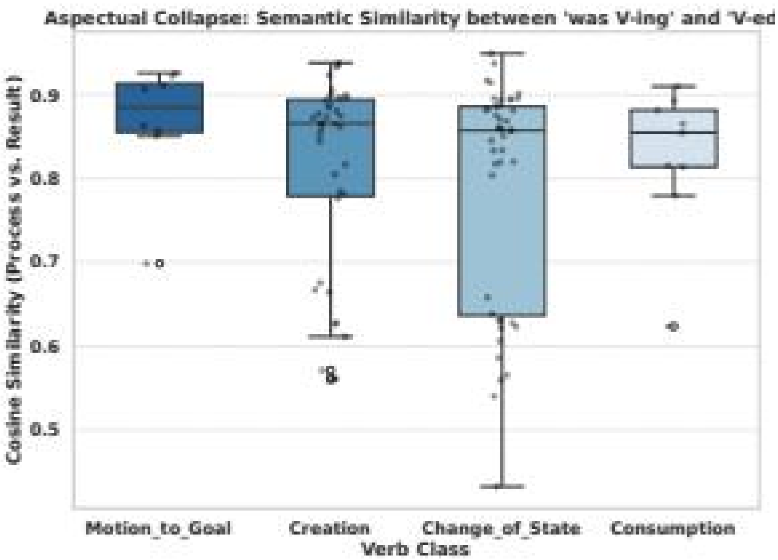

Figure 4: images/5b28d380be32fcb50c457226985da54d5f184ae894ded934e9725e90dce5c36f.jpg

(a) Cosine similarity (Process vs. Result).

(a) 余弦相似度（过程 vs. 结果）。

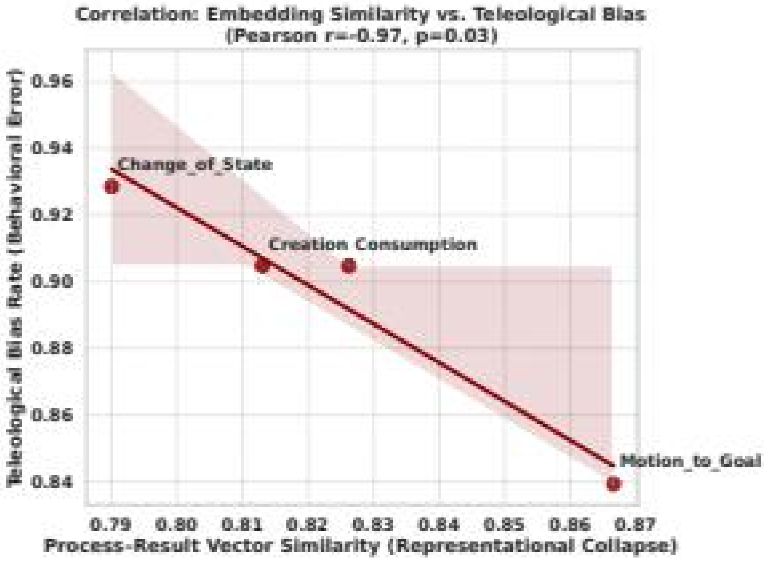

Figure 5: images/91213507c6e51cb966e23f941634595895f435195f40da11a3a800de7acf54b3.jpg

(b) Similarity vs. TBR_C (Zero-shot).
Figure 4: Representational vs. Behavioral Divergence.

(b) 相似性 vs. TBR_C（零样本）。
图 4：表征差异 vs. 行为差异。

tive forms of a verb, and how strongly it hallucinates completion for that verb class. Crucially, this relationship runs in the opposite direction from what a simple confusion account would predict. Motion to Goal verbs show the highest representational similarity (median ≈ 0.88), meaning the model encodes "was flying" and "flew" as nearly identical; yet models reason about them most accurately (with TBR $_C$ being the lowest). Creation verbs show slightly lower similarity (median ≈ 0.85), meaning the model's encoder does distinguish "was building" from "built"; yet these verbs trigger higher hallucination rates (with TBR $_C$ being higher).

动词的不同形式以及模型对该动词类别的完成度产生幻觉的程度。关键的是，这种关系与简单混淆假说所预测的方向相反。运动至目标动词表现出最高的表征相似性（中位数为 ≈ 0.88），这意味着模型将 "正在飞" 和 "飞了" 编码为几乎相同；然而，模型在这些动词上推理的准确性最高（TBR $_C$ 最低）。创造动词表现出稍低的相似性（中位数为 ≈ 0.85），这意味着模型的编码器能够区分 "正在建造" 和 "建造了"；然而这些动词触发了更高的幻觉率（TBR $_C$ 更高）。

This dissociation has a clear implication: the bias does not stem from the model failing to encode the aspectual distinction. If it did, we would expect high similarity to correlate with high error, but the opposite holds. Instead, even when the internal representation correctly separates process from result, the inference step is overridden by a strong prior that Creation events reach their goal. The model "knows" the difference between "was building" and "built", but still predicts completion, because its training has instilled a robust expectation that construction projects succeed. Teleological Bias is therefore not a failure of perception (encoding), but a failure of reasoning (decoding): the model's decoding process is captured by world-knowledge priors rather than guided by the logical content of the input.

这种分离有一个明确的含义：偏差并不是源于模型未能编码事态的区分。如果真的是这样，我们期望高相似度与高错误率相关联，但实际情况却相反。相反，即使内部表示正确地区分了过程和结果，推理步骤仍被一个强烈的先验所覆盖，即创造事件会达到其目标。模型 "知道""正在建造" 和 "建造完成" 之间的区别，但仍预测完成，因为其训练使它形成了一个牢固的期望，即建设项目会成功。因此，目的论偏差并不是感知（编码）的失败，而是推理（解码）的失败：模型的解码过程受到世界知识先验的影响，而非由输入的逻辑内容引导。

6 Conclusion

6 结论

In this work, we evaluated and discovered the Teleological Bias in LLMs using our diagnostic dataset, IMPERFECTIVENLI. Our findings indicate that current models fundamentally struggle to resolve the Imperfective Paradox, frequently conflating the process of a telic event with its culmination. Our investigation into prompting strategies reveals a critical calibration crisis, finding that while some strategies reduce overconfident or hallucinated predictions, they can also make models overly cautious, causing them to reject correct entailments. Meanwhile, this aspectual understanding can improve with model scale unevenly. Furthermore, we demonstrate that this bias is not merely a representational deficit but a decision-time failure: while internal embeddings often distinguish between process and result, strong probabilistic priors regarding goal attainment override these signals during inference. We conclude that solving aspectual reasoning requires moving beyond inference-time prompting toward aligning models to structurally respect the internal logical boundaries of event semantics. We hope that IMPERFECTIVENLI provides a foundation for future work on targeted alignment training, cross-lingual aspectual reasoning, and the broader challenge of teaching LLMs to respect formal semantic boundaries.

在本研究中，我们使用我们诊断数据集 IMPERFECTIVENLI 评估并发现了大语言模型（LLMs）中的目的论偏差。我们的研究结果表明，当前的模型在根本上难以解决非完成体悖论，经常将目的性事件的过程与其结果混淆。我们对提示策略的调查揭示了一个关键的校准危机，发现尽管某些策略可以减少过度自信或幻觉预测，但它们也可能使模型过于谨慎，从而拒绝正确的蕴含关系。同时，这种方面理解能力在模型规模扩大时的提升是不均衡的。此外，我们证明这种偏差不仅仅是一种表征缺陷，而是一种决策时间的失败：虽然内部嵌入通常能够区分过程与结果，但在推理过程中，关于目标达成的强概率先验会覆盖这些信号。我们得出结论，解决方面推理问题需要超越推理时的提示策略，转向使模型在结构上尊重事件语义的内部逻辑边界。我们希望 IMPERFECTIVENLI 能够为未来关于针对性对齐训练、跨语言方面推理以及更广泛的挑战——即教会 LLMs 尊重形式语义边界——的研究提供基础。

7 Limitations

7 局限性

While our work offers important insights into the imperfective paradox with the teleological bias of LLMs, we acknowledge several limitations that define the scope of our findings and point towards future research directions.

尽管我们的研究在具有目标偏见的大型语言模型（LLMs）背景下提供了对未完成体悖论的重要见解，但我们承认一些限制，这些限制界定了我们研究结果的范围，并指出了未来研究的方向。

Ecological Validity and Syntactic Diversity. To strictly isolate the variable of aspectual telicity, we employed a template-based approach to construct IMPERFECTIVENLI. While this ensures high internal validity and precise minimal-pair comparisons, it sacrifices lexical and syntactic diversity. Real-world language usage often includes complex discourse markers, temporal adjuncts (e.g., “for an hour” vs. “in an hour”), and pragmatic cues that constrain interpretation (Zhang, 1995). Consequently, our controlled setup likely overestimates model competence relative to naturalistic settings; handling the imperfective paradox in messier, more complex narratives may prove considerably more challenging.

生态效度和句法多样性。为了严格隔离时态完成性这一变量，我们采用模板化的方法来构建 IMPERFECTIVENLI。虽然这种方法确保了高内部效度和精确的最小对比较，但它牺牲了词汇和句法的多样性。真实语言使用通常包括复杂的语篇标记、时间状语（例如，“持续一小时” 与 “在—小时内”）以及制约理解的语用线索（Zhang, 1995）。因此，我们的受控设置可能相对于自然语境而言高估了模型的能力；处理更加复杂、混乱的叙事中的未完成体悖论可能会更具挑战性。

Linguistic Specificity. Our study focuses exclusively on English, a language that marks the progressive aspect periphrastically (via auxiliary “be” + “-ing”). However, aspect is a typologically diverse category. For instance, Slavic languages mark aspect morphologically (aspect prefixes), while Mandarin Chinese uses aspectual particles. Since LLM tokenizers and representations vary across

语言特异性。我们的研究专门针对英语，这是一种通过助动词 “be” 加上 “-ing” 形式来标记进行体的语言。然而，体是一个类型学上多样化的范畴。例如，斯拉夫语通过形态学方式（体前缀）来标记体，而汉语则使用体标记词。由于大语言模型（LLM）的分词器和表示形式在不同语言中存在差异，因此……

languages, it remains an open question whether the teleological bias is a universal cognitive artifact of Transformers or an artifact of English-centric pre-training data. Meanwhile, in human cognitive studies, research demonstrates that the conceptualization of event semantics differs fundamentally between aspect languages (those with grammaticalized aspect markers, e.g., English, Chinese, Russian) and non-aspect languages (e.g., German) (von Stutterheim et al., 2012; Flecken et al., 2014). Consequently, extending this investigation to a cross-lingual and typological framework represents a vital direction for future research.

语言方面，仍是一个开放的问题：这种目的论偏见是 Transformer 的普遍认知产物，还是以英语为中心的预训练数据的产物。同时，在人类认知研究中，研究表明，事件语义的概念在有方面标记语言（如英语、汉语、俄语等）和无方面标记语言（如德语）之间存在根本性的差异（von Stutterheim 等，2012；Flecken 等，2014）。因此，将这一研究扩展到跨语言和类型学框架中，是未来研究的一个重要方向。

Theoretical Linguistics vs. Human Judgments.

理论语言学 vs. 人类判断。

Since our paper focuses on the theoretical concepts of imperfective paradox, we note potential divergence between theoretical labels and empirical human judgments. As the first study systematically investigating the Imperfective Paradox in LLMs, our primary objective was to characterize the teleological bias, the models’ tendency to hallucinate completion, based on established theoretical semantics. While we focus here on establishing this bias in models, we wish to note that exploring the fine-grained nuances of human perception in these contexts is a vital next step.

由于我们的论文聚焦于不完美性悖论的理论概念，我们注意到理论标签与实证人类判断之间可能存在分歧。作为首项系统研究 LLMs 中不完美性悖论的研究，我们的主要目标是基于已有的理论语义，描述模型的终局性偏见，即模型倾向于虚构完成的倾向。虽然我们在此重点在于确立这种偏见，但我们希望指出，在这些情境下探索人类感知的细微差别是一个至关重要的下一步。

Inference-Only Interventions. Our mitigation strategies were limited to inference-time prompting (CoT, Counterfactuals). While we identified the performance trade-offs as a side effect of these prompts, we did not explore parameter-efficient fine-tuning (PEFT) or activation steering techniques to permanently alter the model’s internal aspectual representations. Future work should investigate whether the “dissociation” between representation and reasoning can be bridged through targeted alignment training rather than transient prompting.

仅推理的干预措施。我们的缓解策略仅限于推理时的提示（如链式推理、反事实方法）。虽然我们识别出这些提示带来的性能权衡是一个副作用，但我们并未探索参数高效微调（PEFT）或激活引导技术，以永久性地改变模型内部的方面表示。未来的工作应探讨是否可以通过有针对性的对齐训练，而非临时提示，来弥合 “表示与推理之间的分离” 这一差距。

Scope of World Knowledge. Finally, our semantic analysis focused on four broad classes of telic verbs. We did not model the granularity of “script knowledge” (Schank and Abelson, 1977; Lyu et al., 2021). For example, the probability of interruption varies by event type (e.g., “building a house” takes months and is prone to interruption, whereas “drawing a circle” takes seconds). Current models may rely on these latent temporal priors, which our uniform evaluation setup does not explicitly disentangle from grammatical aspect.

世界知识的范围。最后，我们的语义分析聚焦于四类广泛的 telic 动词。我们没有对 "脚本知识"（Schank 和 Abelson，1977；Lyu 等，2021）的粒度进行建模。例如，中断的概率因事件类型而异（例如，"建造房屋" 需要数月，容易被打断，而 "画一个圆" 只需几秒钟）。当前的模型可能依赖于这些潜在的时间先验，而我们的统一评估设置并未明确地将这些先验与语法的时态区分开来。

8 Ethical Considerations

8 伦理考量

Human Subjects for Data Evaluation. For the human evaluation of sentence quality, we recruited external participants via the Prolific platform. Prior to the task, annotators were fully informed about the study's scope and provided informed consent. We ensured that no personally identifiable information (PII) was collected. Evaluators were compensated at an hourly rate of £9, exceeding the platform's recommended minimum and comparable to the local living wage.

人类主体用于数据评估。为了对句子质量进行人工评估，我们通过 Prolific 平台招募了外部参与者。在任务开始前，标注人员充分了解了研究的范围并签署了知情同意书。我们确保没有收集任何个人可识别信息（PII）。评估人员的薪酬为每小时 9 英镑，超过了平台推荐的最低标准，并与当地的最低生活工资相当。

Artifact Usage. Regarding the curated artifacts, we release the IMPERFECTIVENLI dataset under the Creative Commons Attribution-NonCommercial 4.0 International License (CC BY-NC 4.0). This license explicitly permits the distribution and use of the dataset for research purposes. We confirm that our use of these existing artifacts is consistent with their intended academic use, and our derivative dataset is released with the strict specification that it be used solely within research contexts, ensuring compatibility with the original access conditions. The dataset contains no harmful or offensive content.

Artifact 使用。关于精选的 artifacts，我们根据知识共享署名-非商业性 4.0 国际许可协议（CC BY-NC 4.0）发布 IMPERFECTIVENLI 数据集。此许可协议明确允许在研究目的下分发和使用该数据集。我们确认，我们使用这些现有 artifacts 的方式与它们的预期学术用途一致，并且我们的衍生数据集发布时严格规定只能在研究环境中使用，以确保与原始访问条件兼容。该数据集不包含任何有害或冒犯性内容。

Use of AI Assistants. As described in Section 3, we used Gemini to assist in the initial generation and curation of the dataset. To ensure data quality and mitigate potential machine artifacts, all generated instances underwent rigorous manual verification and refinement by three native English speakers (see validation details in Section 3 and Appendix C).

使用 AI 助手。如第 3 节所述，我们使用 Gemini 来协助进行数据集的初始生成和整理。为确保数据质量并减少潜在的机器生成痕迹，所有生成的实例均经过三位母语为英语的人员的严格人工验证和优化（详见第 3 节和附录 C 中的验证细节）。

Regarding the manuscript and experimental code, we acknowledge the use of ChatGPT solely for grammatical correction, stylistic polishing, and assistance with preliminary coding routines. The authors verified all AI-assisted outputs and bear full responsibility for the accuracy and originality of the content presented in this work.

关于手稿和实验代码，我们承认仅使用 ChatGPT 进行语法纠正、风格润色以及协助初步编码工作。作者验证了所有人工智能辅助输出的内容，并对本文中呈现内容的准确性和原创性承担全部责任。

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